O-P-921c

May 20, 1963

SUPERSEDING
Int. Fed. Spec. 00921b (GSA-FSS)
September 29, 1961 and
Fed. Spec. O-P-921a
June 11, 1953

FEDERAL SPECIFICATION

# PYRETHRUM EXTRACT

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE

1.1 This specification covers extract obtained from authentic pyrethrum flowers upon treatment with an appropriate solvent or solvents, suitable for compounding of aerosol insecticides, and hereinafter called pyrethrum extract.

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

*Federal Specifications:*

P-S-616—Soap, Toilet (Floating, White).

BB-C-310—Chlorofluoro Hydrocarbons (Of the Methane and Ethane Series).

VV-K-220 — Kerosene, Deodorized

PPP-D-729—Drums, Metal, 55-Gallon (For Shipment of Noncorrosive Material).

*Federal Standards:*

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

*Military Standards:*

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

FSC 6840



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(Copies of Military specifications and standards, required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

*Governmental*:

Code of Federal Regulations, Title 49, Transportation, Chapter 1, Interstate Commerce Commission, Parts 71-78; Explosives and Other Dangerous Articles—49 CFR 71-78.

U. S. Coast Guard Regulations for Explosives and Other Dangerous Articles on Board Vessels.

Federal Insecticide, Fungicide, and Rodenticide Act.

(Copies may be obtained upon application, accompanied by money order, coupon or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

*Nongovernmental*:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Official Methods of Analysis of the Association of Official Agricultural Chemists are published by the Association of Official Agricultural Chemists, P. O. Box 540, Benjamin Franklin Station, Washington 4, D. C.)

Chemical Specialties Manufacturers Association;

Method No. 23—Insoluble Matter in Dichlorodifluoromethane of Pyrethrum Extract.

(Copies are available from the Chemical Specialties Manufacturers Association, 50 East 41st Street, New York 17, New York.)

## 3. REQUIREMENTS

3.1 **Material.** The extract shall be composed of naturally occurring substances extracted from commercial pyrethrum flowers and dissolved in a water-white, odorless refined kerosene. No other compounds shall be added as solvents or for such purposes as extenders, perfumes, synergists, or activators. Anti-oxidants are permitted.

3.2 **Pyrethrins content.** The pyrethrum extract shall have a nominal pyrethrins content of 20 percent by weight when tested as specified in 4.4.1. Subject to adjustment between supplier and purchaser, material containing between 17 and 23 percent pyrethrins by weight may be accepted.

3.3 **Corrosion.** The extract shall have no detrimental action upon metal or paint when tested in accordance with 4.4.2.

3.4 **Neutralization number.** The neutralization number shall be not less than 5 nor more than 20 when determined by the procedure specified in 4.4.3.

3.5 **Moisture.** The moisture content shall not exceed 0.10 percent when determined by the procedure specified in 4.4.4.

3.6 **Absence of staining.** Pyrethrum extract shall be nonstaining when tested as specified in 4.4.5.

3.7 **Insoluble matter.** The matter insoluble in dichlorodifluoromethane shall not exceed 0.5 percent and shall be determined by the procedure specified in 4.4.6.

3.8 **Labeling.** The material shall be labeled in compliance with the requirements of the Federal Insecticide, Fungicide, and Rodenticide Act.

3.9 **Workmanship.** The product shall be clean, uniform, and free from any defects which might impair its utility.



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## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Sampling lot.* For purposes of sampling, a lot shall consist of material from one manufactured batch, offered for delivery at one time.

4.1.2 *Sampling procedure for lot acceptance tests.* From each lot, at least two separate 1-pint sample specimens shall be taken at random from two drums in the lot so that a measure of variability within each lot may be obtained. The samples shall be placed in clean, dry, lightproof or amber glass containers which shall be tightly sealed and carefully marked. The thiefs and bottles for sampling shall be cleaned carefully with trichloroethylene and dried before using. Sample bottles shall be kept corked before and after samples have been taken. The samples shall be forwarded to the laboratory for the tests specified in 4.4.

4.1.3 *Sampling for inspection of containers.* A random sample of filled containers shall be taken in accordance with MIL-STD-105 at inspection level I or as otherwise specified in the contract or order (see 6.2), and acceptable quality level = 2.5 percent defective to verify compliance with all requirements regarding fill, closure, marking, and other requirements not requiring tests.

**4.2 Inspection.** The materials, which do not require tests, shall be inspected to determine conformance with the requirements of this specification.

**4.2.1** ***Inspection of containers.*** **Each sample filled container selected in accordance with 4.1.3 shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.**

**4.3 Lot acceptance tests.**

**4.3.1** ***Procedure.*** **The portion of the sample selected and forwarded as specified in 4.1.2 shall be subjected to the tests specified in 4.4. The sample shall be tested within 30 days from the time it is taken. If the sample is found to be not in conformance with this specification, the entire lot represented shall be rejected.**

**4.4 Test procedures.**

**4.4.1** ***Pyrethrins content.*** **The pyrethrins content shall be determined in accordance with the mercury reduction method described in 4.107 of the Official Methods of Analysis of the Association of Official Agricultural Chemists.**

**4.4.2** ***Corrosion.*** **The extract shall be mixed with dichlorodifluoromethane conforming to type 12 of BB-C-310 in the ratio of eight parts by weight of extract to one hundred and fifty-two parts by weight of dichlorodifluoromethane. Spray one-half of the surface of clean test panels of uniform shape and size with the aerosol produced from the above mixture. Hold the spraying nozzle approximately 6 inches away from the panels and spray for one minute. The**



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panels shall be one each of carbon steel, brass, copper, baked enamel on steel, varnish on steel, and lacquer on steel. After spraying, allow panels to stand for 1 hour in a room at a temperature of 70° to 80° F. Then wash with a water solution of soap conforming to P-S-616. Permit panels to dry; inspect for any detrimental action.

4.4.3 *Neutralization number.* A 1- to 2-gram sample of the extract is weighed accurately into a 250-ml. Erlenmeyer flask and 50 ml. of a freshly prepared, neutralized 50-50 mixture of ethyl alcohol and ethyl ether is added. Add 3 or 4 drops of phenolphthalein solution and titrate with 0.02 *N* aqueous alkali, until a pink color persists for 60 seconds. Calculate the neutralization number as mg. KOH per gram of sample.

4.4.4 *Moisture content.* Moisture shall be determined by titration with a solution of iodine, sulfur dioxide and pyridine in methanol (Karl Fischer Reagent). The procedure shall be as specified in method 4082 of Fed. Test Method Std. No. 141 with the substitution of pyrethrum extract for paint or varnish.

4.4.5 *Absence of staining.* To 0.1 ml. of pyrethrum extract add sufficient water-white kerosene conforming to VV-K-220 to make 10 ml. of solution. Apply prepared solution to bleached, unsized white cotton cloth or white filter paper, and allow to evaporate. No colored or oily stain shall remain after evaporation.

4.4.6 *Insoluble matter.* Insoluble matter shall be determined in accordance with method No. 23 of the Insecticide Committee of the Chemical Specialties Manufacturers Association.

## 5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packing.**

5.1.1 *Levels A and B.* The material shall be furnished in 55-gallon drums conforming to type I of PPP-D-729. Interior of the drum shall be lined with tin, epoxy resin, or baked phenolic resin. Exterior of the drum shall be coated with a rust inhibiting enamel in accordance with paragraph 3.1.3 of the drum specification.

5.1.2 *Level C.* Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to the marking specified in 3.8 and any special markings required by the contract or order, marking shall be in accordance with Fed. Std. No. 123. In addition, shipping containers shall be marked to insure safe handling as required by the Code of Federal Regulations and Regulations for Explosives or Other Dangerous Articles on Board Vessels.

5.2.2 *Military agencies.* In addition to the marking specified in 3.8 and any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129. In addition, shipping containers shall be marked to insure safe handling as required by the Code of Federal Regulations and Regulations for Explosives or Other Dangerous Articles on Board Vessels.

## 6. NOTES

6.1 **Intended use.** Pyrethrum extract is intended for use as an insecticide ingredient in aerosol insecticides.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.



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(*b*) Inspection level required if other than that specified in 4.1.3.

(*c*) Level of packing required (see 5.1).

6.3 Pyrethrum extract should be purchased by net weight.

6.4 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

*Rail*:

Insecticides, not otherwise indexed by name, other than agricultural.
Carload minimum weight 36,000 pounds.

*Motor*:

Insecticides, not otherwise indexed, other than agricultural.
Truckload minimum weight 36,000 pounds, subject to Rule 115 National Motor Freight Classification.

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